Symplectic groupoids via 2-shifted lagrangians.

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Shifted symplectic structures for the working Poisson geometer

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Central object of study: **Symplectic groupoids**

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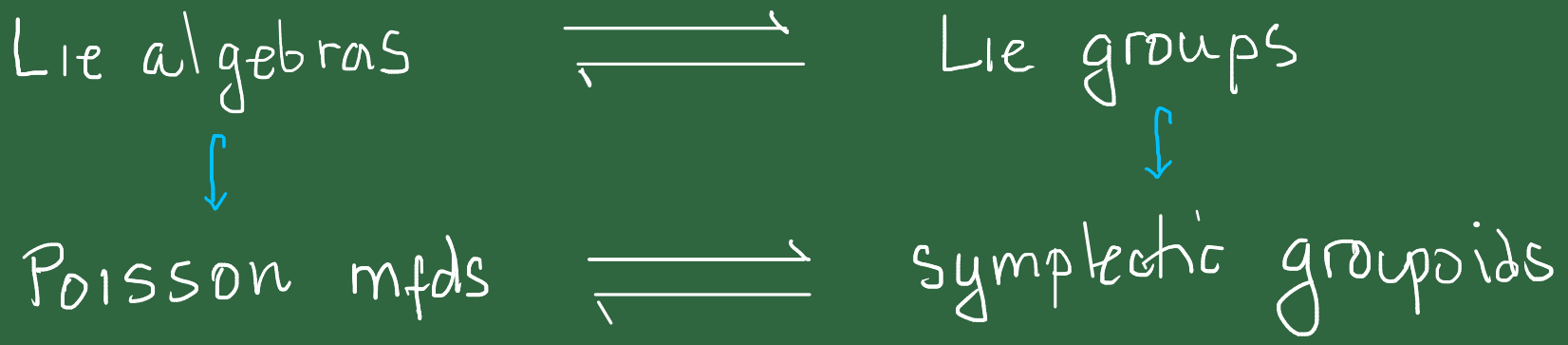
Central object of study: **Symplectic groupoids**

Many reasons:
- quantization
- hamiltonian actions, Morita equivalence
- Poisson invariants, Linearization, PMcT

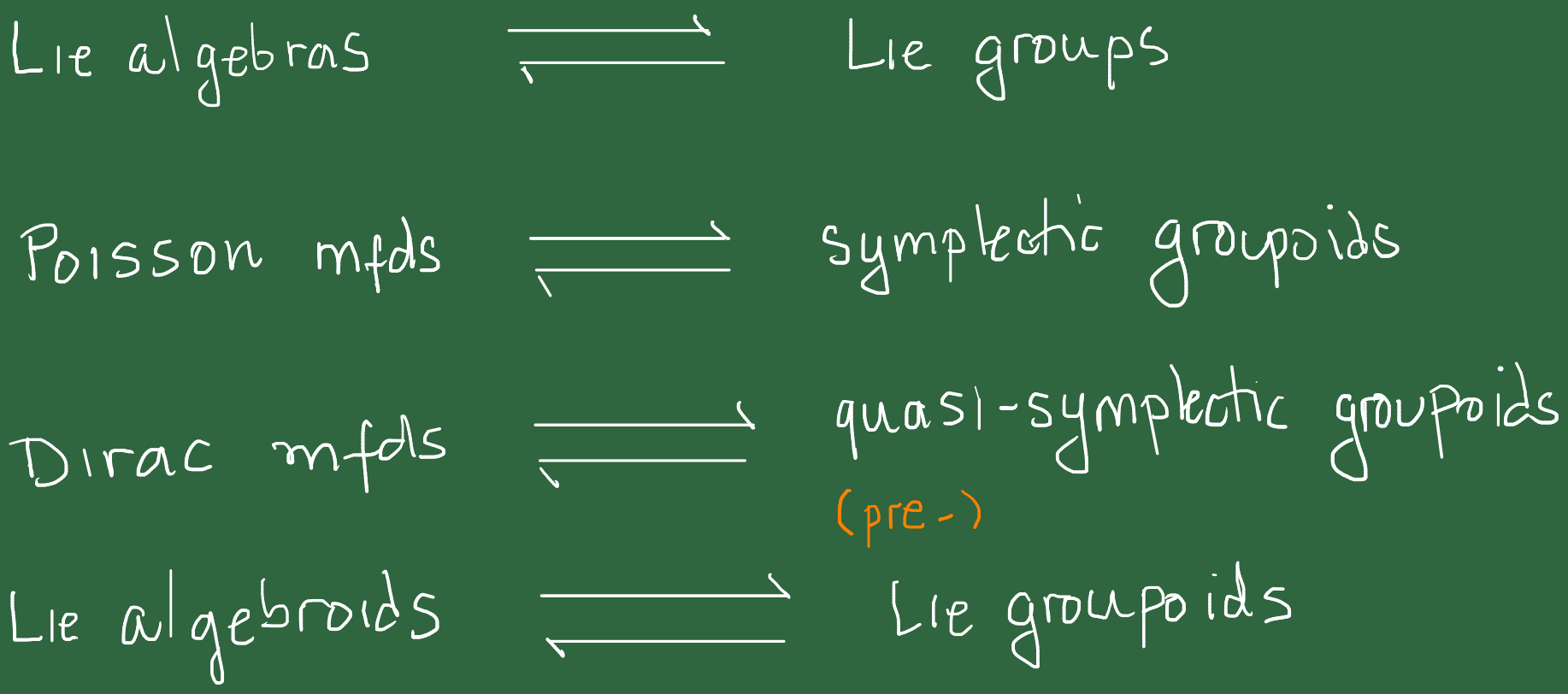
general context:

Poisson mfds $\underset{\text{duft.}}{\overset{\text{integ.}}{\rightleftharpoons}}$ symplectic groupoids

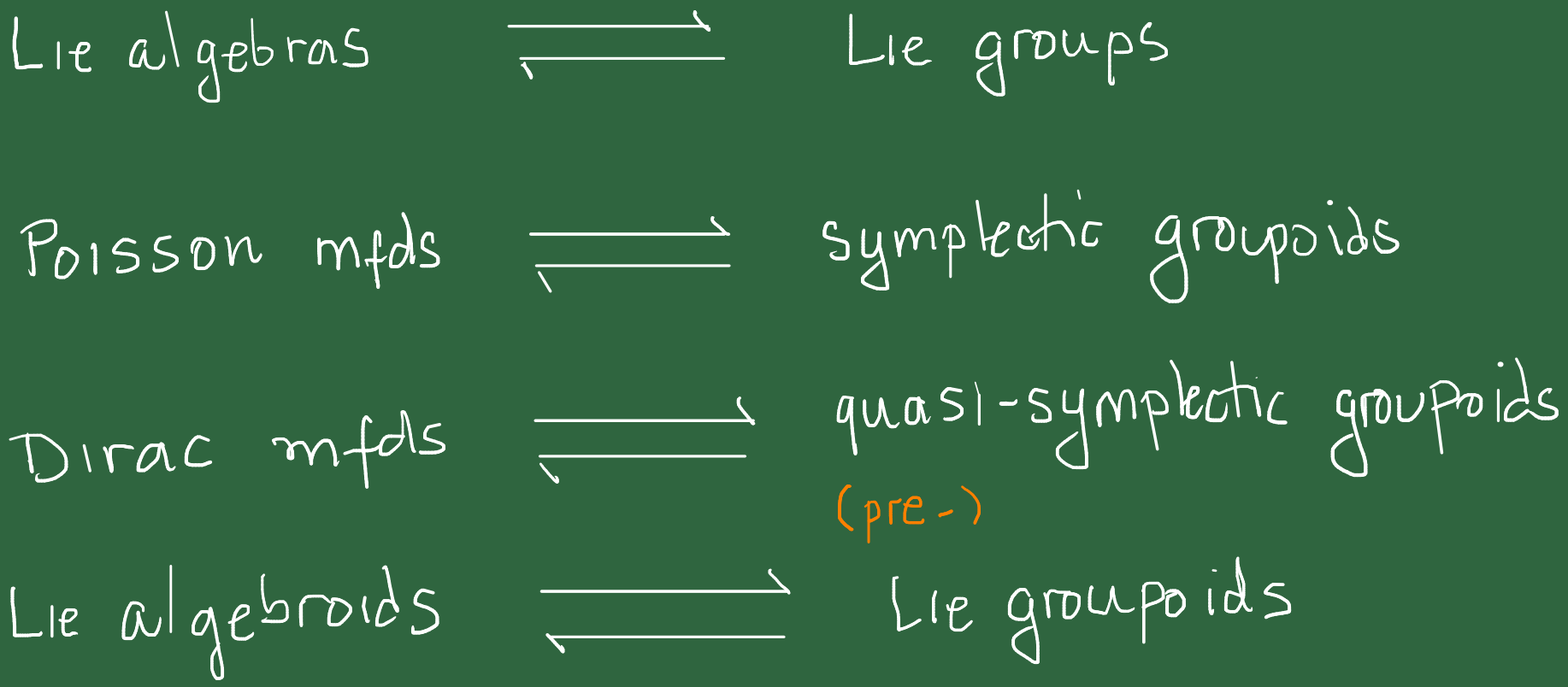
general context:



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But NOT every Poisson manifold is integrable...

Many important classes of Poisson str. are integrable,

Eg symplectic, Poisson Lie groups, quotients M/G, bg-symplectic...

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But still not obvious how to describle symplectic groupoids...

Goal:
- shed light on constructions of (quasi-) symplectic groupoids
- Shifted symplectic geometry as tool

(quasi-) symplectic groupoids $\subseteq$ "2-shifted lagrangian structs"

fibred products

Outline :

- "General" framework { m-shifted symplectic groupoids $\mathcal{G}$, ω. ($\mathcal{G} \twoheadrightarrow M$)
  Lagrangian structures on $\mathcal{H} \xrightarrow{\Phi} \mathcal{G}$ ($\mathcal{H} \twoheadrightarrow N$, $\mathcal{G} \twoheadrightarrow M$)

Outline:

- "General" framework { m-shifted symplectic groupoids $\mathcal{G}$, ω. ↓ M
  Lagrangian structures on $\mathcal{H} \xrightarrow{\Phi} \mathcal{G}$, $\mathcal{H}$ ↓ N, $\mathcal{G}$ ↓ M
- 0, 1 shifts
- 2-shifted story

  UI

  quasi-Poisson manifolds

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  ⇕

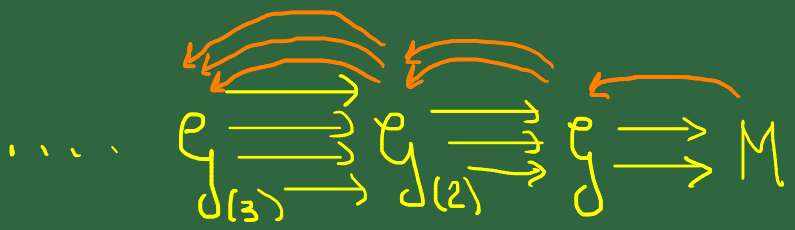
  quasi-Poisson manifolds
- Application: integration of affine Dirac structures, symplectic groupoids of Poisson homogeneous spaces $\mathcal{G}$ ⇊ G/H

  (B., Iglesias, Lu; JDG '24)

① "General" framework (Getzler, Calaque, Safronov ...)

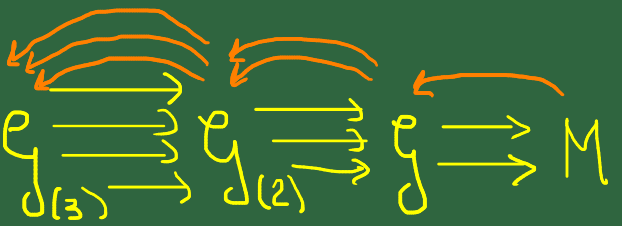
① "General" framework

$\mathcal{G} \underset{t}{\overset{s}{\rightrightarrows}} M$ Lie groupoid



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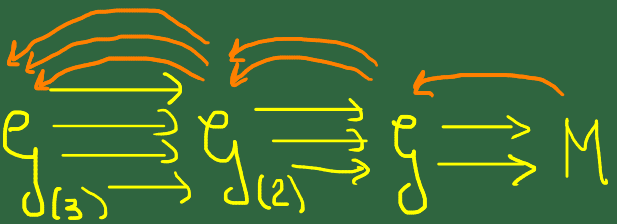


Bott-Shulman-Stasheff : $(\Omega^{\cdot}(\mathcal{G}_{\cdot}), \delta, d)$

$$
\begin{array}{ccccccc}
\vdots & & \vdots & & \vdots & & \\
\uparrow & & \uparrow & & \uparrow & & \\
\Omega^2(M) & \longrightarrow & \Omega^2(\mathcal{G}) & \longrightarrow & \Omega^2(\mathcal{G}_{(2)}) & \longrightarrow & \cdots \\
\uparrow & & \uparrow & & \uparrow & & \\
\Omega^1(M) & \longrightarrow & \Omega^1(\mathcal{G}) & \longrightarrow & \Omega^1(\mathcal{G}_{(2)}) & \longrightarrow & \cdots \\
\uparrow d & & \uparrow & & \uparrow & & \\
C^\infty(M) & \xrightarrow{\delta} & C^\infty(\mathcal{G}) & \longrightarrow & C^\infty(\mathcal{G}_{(2)}) & \longrightarrow & \cdots
\end{array}
$$

① "General" framework

$\mathcal{G} \rightrightarrows M$ Lie groupoid



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$$

$\hat{\Omega}^{\cdot}(\mathcal{G}_{\cdot})$ normalized subcomplex

$\hat{\Omega}^{\cdot}(\mathcal{G}_{\cdot})$ —VAN EST→ $W^{\cdot}(A)$

m-shifted n-form on $\mathcal{G} \rightrightarrows M$:

$$\omega_\bullet = \omega_m + \omega_{m-1} + \cdots + \omega_1 + \omega_0$$

$$\hat{\Omega}^n(\mathcal{G}_{(m)}) \quad \hat{\Omega}^{n+1}(\mathcal{G}_{(m-1)}) \qquad \hat{\Omega}^{n+m-1}(\mathcal{G}) \quad \Omega^{n+m}(M)$$

$n=2$:

- 0-shifted $\alpha \in \Omega^2(M)$
- 1-shifted $\omega \in \Omega^2(\mathcal{G})$, $H \in \Omega^3(M)$
- 2-shifted $\Omega \in \Omega^2(\mathcal{G}_{(2)})$, $\theta \in \Omega^3(\mathcal{G})$, $H \in \Omega^4(M)$

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Closed m-shifted 2-form : $\mathcal{D}\omega_\bullet = 0$

$\mathcal{D}$ = total differential

$\mathcal{D}\omega_\bullet$ is $(m+1)$-shifted 2-form

non degenerate? " $\omega_\bullet^\flat : T\mathcal{G} \xrightarrow{\sim} T^*\mathcal{G}$ "

Tangent complex of $\mathcal{G}$: $\mathbb{T}_{\mathcal{G}} = \left(\underset{1}{A} \xrightarrow{\beta} \underset{0}{TM} \right)$

ω. closed m-shifted 2-form: $\Lambda_{\omega_m} : \mathbb{T}_{\mathcal{G}} \longrightarrow \mathbb{T}^*_{\mathcal{G}}[-m]$ chain map

Tangent complex of $\mathcal{G}$: $\mathcal{T}_{\mathcal{G}} = \left(A \xrightarrow{\rho} TM \right)$ (degrees: A in degree 1, TM in degree 0)

ω_m closed m-shifted 2-form: $\Lambda_{\omega_m} : \mathcal{T}_{\mathcal{G}} \longrightarrow \mathcal{T}^*_{\mathcal{G}}[-m]$ charn map

nondegenerate: Λ_{ω_m} is quasi-isomorphism

0-shifted

$$\begin{array}{ccccc} A & \xrightarrow{\rho} & TM & \longrightarrow & 0 \\ & & \downarrow \alpha & & \\ 0 & \longrightarrow & T^*M & \xrightarrow[\rho^*]{} & A \end{array}$$

1-shifted

$$\begin{array}{ccc} A & \xrightarrow{\rho} & TM \\ \downarrow \lambda_\omega & & \downarrow -\lambda^*_\omega \\ T^*M & \longrightarrow & A^* \end{array}$$

2-shifted

$$\begin{array}{ccccc} 0 & \longrightarrow & A & \xrightarrow{\rho} & TM \\ & & \downarrow \lambda_{\omega_2} & & \\ T^*M & \xrightarrow[\rho^*]{} & A^* & \longrightarrow & 0 \end{array}$$

- $(\mathcal{G}, \omega_\bullet)$ n-shifted symplectic groupoid (closed + nondeg.)

- $$\begin{array}{ccc} \mathcal{H} & \xrightarrow{\Phi} & \mathcal{G} \\ \downarrow\downarrow & & \downarrow\downarrow \\ N & \xrightarrow{\varphi} & M \end{array}$$ morphism

lagrangian ?

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lagrangian ?

classically: $\varphi : N \hookrightarrow M, \omega$

- $\varphi^* \omega = 0$ (isotropic)

- $\mathcal{V}_N = \dfrac{\varphi^* TM}{TN} \xrightarrow{\sim} T^*N$

$\omega^\flat$: $\varphi^* TM \to T^*N$; $d\varphi^*$: $T^*M \to \varphi^* TM$

- $(\mathcal{G}, \omega_\bullet)$ m-shifted symplectic groupoid (closed + nondeg.)

- $$\begin{array}{ccc} \mathcal{H} & \xrightarrow{\Phi} & \mathcal{G} \\ \downarrow\downarrow & & \downarrow\downarrow \\ N & \xrightarrow{\varphi} & M \end{array}$$ morphism

- <u>isotropic structure on Φ</u> : $\beta_\bullet$ $(m-1)$-shifted 2-form on $\mathcal{H}$,

$$\boxed{D\beta_\bullet = \Phi^t \omega_\bullet}$$

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- Lagrangian condition :

$$\Lambda_{\Phi,\beta} : \mathcal{L}_\Phi \xrightarrow{\sim} \tau^*_{\mathcal{H}}[1-m] \quad \text{quasi-isom.}$$

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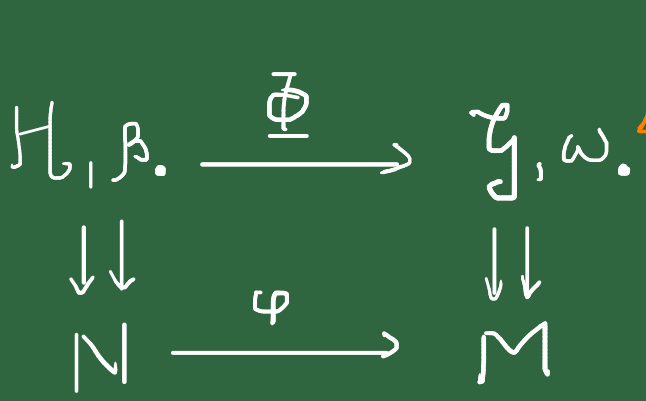
$$\boxed{\Lambda_{\Phi,\beta} : \mathcal{V}_\Phi \xrightarrow{\sim} \mathbb{T}_{\mathcal{H}}^*[1-m] \quad \text{quasi-isom.}}$$

$$\mathcal{V}_\Phi = \left(A_{\mathcal{H}} \xrightarrow{(\rho_{\mathcal{H}}, Lie(\Phi))} \mathbb{T}N \oplus \varphi^* A_{\mathcal{G}} \xrightarrow{\tau_\varphi - \rho_{\mathcal{G}}} \varphi^* \mathbb{T}M \right)$$

« normal bundle »
(mapping cone of $\tau\Phi : \tau_{\mathcal{H}} \to \tau_{\mathcal{G}}$)

$$\Lambda_{\Phi,\beta}(v \oplus w) = \Lambda_{\beta_{m-1}}(v) + (\tau\Phi)^*(\Lambda_{\omega_m}(w))$$

$\tau_{\mathcal{H}}$ $\quad \varphi^* \tau_{\mathcal{G}}[1]$



$\beta_\bullet$ Lagrangian structure $\begin{cases} D\beta_\bullet = \Phi^* \omega_\bullet \\ \mathcal{V}_\Phi \xrightarrow{\sim} \mathcal{T}^\chi_{\mathcal{H}}[1-m] \end{cases}$

Property: $\mathcal{H}, \beta_\bullet \longrightarrow \{*\}$ is $(m-1)$ shifted symplectic groupoid

$$\begin{array}{ccc} \mathcal{H}, \beta_\bullet & \xrightarrow{\Phi} & \mathcal{G}, \omega_\bullet \\ \downarrow\downarrow & & \downarrow\downarrow \\ N & \xrightarrow{\varphi} & M \end{array}$$

m-shifted symplectic

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Thm:

$$\begin{array}{ccc} \mathcal{H}_1, \beta_1 & \xrightarrow{\Phi_1} & \\ & & (\mathcal{G}, \omega) \\ \mathcal{H}_2, \beta_2 & \xrightarrow{\Phi_2} & \end{array} \qquad \Phi_1 \pitchfork \Phi_2$$

m-shifted symplectic

Lagrangians

$$\Longrightarrow \quad \mathcal{H}_1 \underset{\mathcal{G}}{\times} \mathcal{H}_2, \quad \beta = pr_1^* \beta_1 - pr_2^* \beta_2$$

is $(m-1)$ shifted symplectic

② 0-shifted case $\mathcal{G} \rightrightarrows M$

- $\alpha \in \Omega^2(M)$; $d\alpha = 0$, $s^*\alpha = t^*\alpha$ (closed)
- ρ injective , $\ker(\alpha) = \rho(A)$ (nondeg.)

Foliation groupoid, $\alpha \in \Omega^2(M)$ closed, basic, transversally nondegenerate

Eg $G_\xi \ltimes \bar{\mu}^{-1}(\xi) \rightrightarrows \bar{\mu}^{-1}(\xi)$ $\mu : M \to \mathfrak{g}^*$ moment map

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$\mathcal{G}$ foliation groupoid, $\alpha \in \Omega^2(M)$ closed, basic, transversally nondegenerate

Eg $G_\xi \ltimes \bar{\mu}^{-1}(\xi) \rightrightarrows \bar{\mu}^{-1}(\xi)$

- $\begin{array}{ccc} \mathcal{H} & \xrightarrow{\Phi} & \mathcal{G} \\ \downarrow\downarrow & & \downarrow\downarrow \\ N, \beta=0 & & M, \alpha \end{array}$ Lagrangian $\iff$ $\rho_{\mathcal{H}}$ injective

$$\frac{TN}{\rho_{\mathcal{H}}(A_{\mathcal{H}})} \hookrightarrow \varphi^* \frac{TM}{\rho(A)}$$

injective Lagrangian

1-shifted case

$\mathcal{G} \rightrightarrows M$

- $\omega \in \Omega^2(\mathcal{G})$, $H \in \Omega^3(M)$;

  ω multiplicative, $d\omega = s^*H - t^*H$, $dH = 0$ (closed)

- $$\begin{array}{ccc} A & \xrightarrow{\rho} & TM \\ {\scriptstyle \lambda_\omega}\downarrow & & \downarrow{\scriptstyle -\lambda_\omega} \\ T^*M & \xrightarrow{\rho^*} & A^* \end{array}$$

  quasi-isom. $\iff$ $\begin{cases} \ker(\omega) \cap \ker(ds) \cap \ker(dt) = \{0\} \\ \dim \mathcal{G} = 2 \dim M \end{cases}$

  BCWZ, X_u

$\iff$ $\omega : T\mathcal{G} \longrightarrow T^*\mathcal{G}$ Morita morphism

delHoyo-Ortiz

1-shifted case $\quad \mathcal{G} \rightrightarrows M$

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1-shifted symplectic groupoid = quasi-symplectic groupoid

- infinitesimally: (BCWZ)

  quasi-sympl. grp $\mathcal{G}$, $\omega + H$ $\rightleftharpoons$ Dirac structures $L \subseteq \mathbb{T}_H M$

Lagrangian

$$\begin{array}{ccc} \mathcal{H} & \xrightarrow{\Phi} & \mathcal{G} \quad \omega \\ \downarrow\downarrow & & \downarrow\downarrow \\ N & \xrightarrow{\varphi} & M \quad H \end{array}$$

- $\beta \in \Omega^2(N)$ $\quad \begin{cases} s^*\beta - t^*\beta = \Phi^*\omega \\ d\beta = \varphi^* H \end{cases}$
- $A_{\mathcal{H}} \underset{\uparrow}{\simeq} \left\{ (Y, (X, \alpha)) \in TN \oplus \overset{\leftarrow}{\varphi} L \; ; \; \begin{array}{l} d\varphi(Y) = X \\ i_Y \beta = -\varphi^* \alpha \end{array} \right\}$

$(\rho_{\mathcal{H}}, \mathrm{Lie}(\Phi))$

Lagrangian

$$\begin{array}{ccc} H & \xrightarrow{\Phi} & \mathcal{G} \quad \omega \\ \downarrow\downarrow & & \downarrow\downarrow \\ N & \xrightarrow{\varphi} & M \quad H \end{array}$$

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$(\rho_H, \mathrm{Lie}(\Phi))$

(Crooks-Mayrand)

special cases :

- $H \hookrightarrow \mathcal{G}$ Lie subgroup — "stabilizer subgroupoids" $\quad \mathcal{G}_x \hookrightarrow \mathcal{G}$

eg Hamiltonian G-space $\quad \begin{array}{c} T^*G \\ \downarrow\downarrow \\ \mathfrak{g}^* \end{array}$

quasi hamiltonian $\quad \begin{array}{c} G\times G \\ \downarrow\downarrow \\ G \end{array}$

- $\begin{array}{ccc} \mathcal{G}\times N & \dashrightarrow & \mathcal{G} \\ \downarrow\downarrow & & \downarrow\downarrow \\ N & \xrightarrow{\mu} & M \end{array}$ — "Hamiltonian spaces" $\quad (N,\beta) \longrightarrow (M,L)$

strong Dirac map

Lagrangian

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$(\rho_{\mathcal{H}}, \mathrm{Lie}(\Phi))$

special cases :

- $H \hookrightarrow \mathcal{G}$ Lie subgroup — "stabilizer subgroupoids" $\mathcal{G}_x \hookrightarrow \mathcal{G}$
- $\begin{array}{ccc} \mathcal{G}\times N & \dashrightarrow & \mathcal{G} \\ \downarrow\downarrow & & \downarrow\downarrow \\ N & \xrightarrow{\mu} & M \end{array}$ — "Hamiltonian spaces" $(N,\beta) \longrightarrow (M,L)$ strong Dirac map

Lagrangian product = Reduction

$$\begin{array}{ccccc} & & \mathcal{G}_x \propto \mu^{-1}(x) & & \\ & \swarrow & & \searrow & \\ \mathcal{G}\times N & & & & \mathcal{G}_x \\ \downarrow\downarrow & \searrow & & \swarrow & \downarrow\downarrow \\ N & & \mathcal{G} & & \{x\} \\ & \searrow & \downarrow\downarrow & \swarrow & \\ & & M & & \end{array}$$

③ The 2-shifted case

$\mathcal{G} \rightrightarrows M$ $\quad \omega_2 + \omega_1 + \omega_0$

$\Omega^2(\mathcal{G}_{(2)})$ $\Omega^3(\mathcal{G})$ $\Omega^4(M)$

- nondegeneracy $\Rightarrow$ $\mathcal{G}$ transitive

  $\Rightarrow$ Morita equiv to Lie group

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The example we care about:

$\mathfrak{g}, \langle \cdot, \cdot \rangle$ quadratic Lie algebra ; D Lie group

$$\Omega = -\frac{1}{2} \langle pr_1^* \theta^L, pr_2^* \theta^R \rangle \in \Omega^2(D \times D)$$

$$\theta = \frac{1}{12} \langle \theta^L, [\theta^L, \theta^L] \rangle \in \Omega^3(D)$$

$(D, \Omega + \theta)$ is 2-shifted symplectic Lie group $\overset{\text{V.E.}}{\rightsquigarrow}$ $(\mathfrak{g}, \langle \cdot, \cdot \rangle)$

diff

Lagrangian

$$\begin{array}{ccc} \mathcal{H} & \xrightarrow{\Phi} & D \\ \downarrow\downarrow & & \downarrow\downarrow \\ N & & \{*\} \end{array}$$

I) $\beta \in \Omega^2(\mathcal{H})$ $\quad \varepsilon^*\beta = 0$

$H \in \Omega^3(N)$

$$\begin{cases} pr_1^*\beta + pr_2^*\beta - m^*\beta = \Phi_{(2)}^* \Omega \\ s^* H - t^* H - d\beta = \Phi^* \theta \end{cases}$$

$dH = 0$

DIRAC

$$A_{\mathcal{H}} \xrightarrow{\sim} L \subseteq T_H N \times \mathfrak{A}$$

$(\rho_H, \lambda_\beta), \mathrm{Lie}(\Phi)$

↖ product Courant algb.

Lagrangian

$$\begin{array}{ccc} \mathcal{H} & \xrightarrow{\Phi} & D \\ \downarrow\downarrow & & \downarrow\downarrow \\ N & & \{*\} \end{array}$$

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  $H \in \Omega^3(N)$

$$(*)\begin{cases} pr_1^*\beta + pr_2^*\beta - m^*\beta = \Phi_{(2)}^*\Omega \\ s^*H - t^*H - d\beta = \Phi^*\theta \end{cases} \qquad dH = 0$$

- $A_{\mathcal{H}} \xrightarrow{\sim} L \subseteq T_H N \times \mathcal{J}$ (DIRAC)

  $(\rho_H, \lambda_\beta), \mathrm{Lie}(\Phi)$ ← product Courant algb.

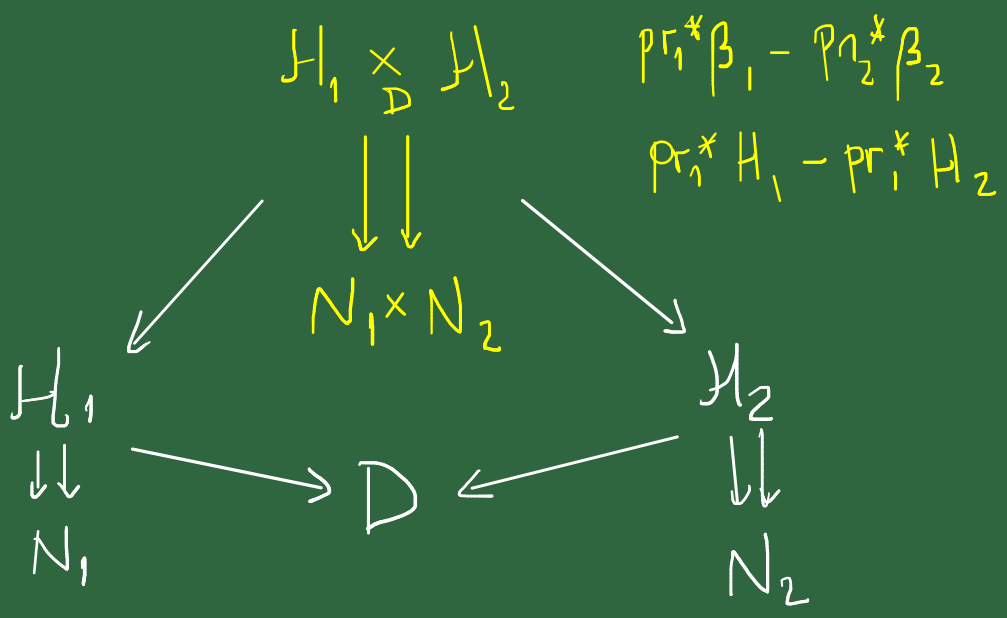
Thm:

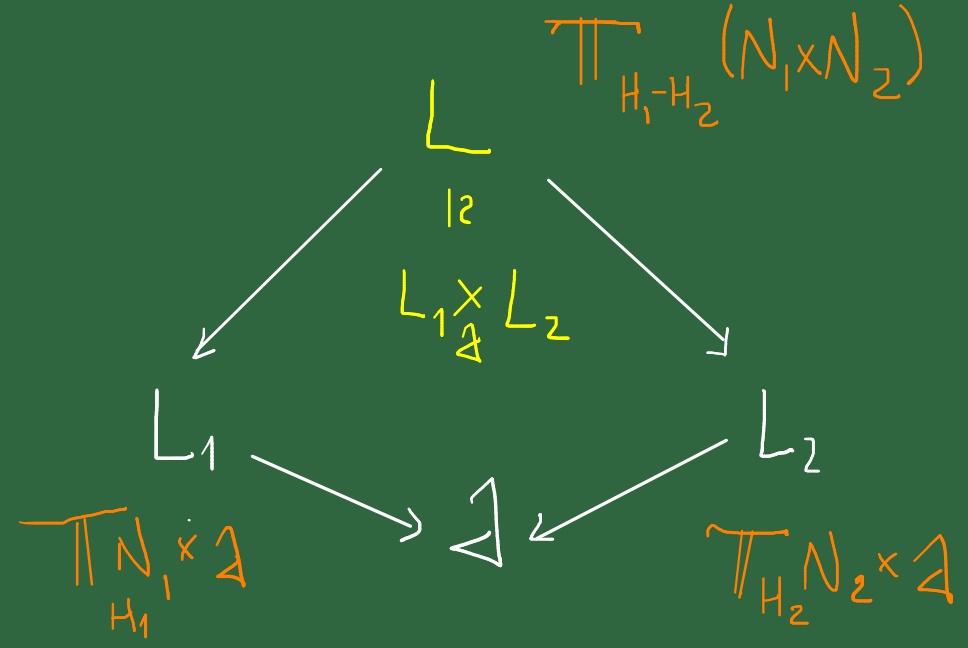
- β, H is Lagrangian iff $(*)$ and
$$\begin{cases} \dim \mathcal{H} - 2\dim N = \frac{1}{2}\dim D \\ \ker d\phi \cap \ker\beta \cap \ker ds \cap \ker dt = \{0\} \end{cases}$$

- $\mathcal{H} \xrightarrow{\Phi} D$ Lagrangian $\underset{\text{int}}{\overset{\text{diff}}{\rightleftarrows}}$ $L \subseteq T_H N \times \mathcal{J}$ Dirac structure

(BCWZ as special case...)

quasi-symplectic groupoids as 2-shifted Lagrangian "intersections"





Many examples of 2-shifted lagrangians in Poisson geometry!

(quasi-Poisson actions

in summary:

Poisson mfds ↕ $L \subseteq T_H M$ Dirac ⟺ symplectic grpds ↕ quasi-symplectic $\mathcal{G} \rightrightarrows M$ ↕

$L \subseteq T_H M \times \lambda$ Dirac ⟺ lagrangian $\mathcal{G} \xrightarrow{\Phi} D$

$$pr_1^* \omega + pr_2^* \omega - m^* \omega = \Phi_2^* \Omega \qquad dH = 0$$

$$s^* H - t^* H - d\omega = \Phi^* \theta$$

$$\dim \mathcal{G} - 2 \dim M = \tfrac{1}{2} \dim D$$

$$\ker d\Phi \cap \ker \omega \cap \ker ds \cap \ker dt = \{0\}$$

quasi-Poisson

- M manifold, $\pi \in \mathfrak{X}^2(M)$
- $\mathfrak{g}$, $[\cdot,\cdot]$, $F : \mathfrak{g} \to \wedge^2 \mathfrak{g}$, $\chi \in \wedge^3 \mathfrak{g}$
- $\mathfrak{g} \xrightarrow{\rho_M} \mathfrak{X}(M)$ action

Lie quasi-bialgebra

quasi Manin triple

$\mathfrak{d} = \mathfrak{g} \oplus \mathfrak{g}^*$

quasi-Poisson

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$$\begin{cases} \mathcal{L}_{\rho_M(u)} \pi = -\rho_M(F(u)) \\ \frac{1}{2}[\pi,\pi] = \rho_M(\chi) \end{cases}$$

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Another viewpt:

$$T^*M \times \mathfrak{g} \xrightarrow{\Delta} TM \times \mathfrak{g}^*$$
$$(u, \alpha) \longmapsto (\pi^\#(\alpha) + \rho_M(u), \rho_M^* \alpha)$$

B., Crainic, Ševera

$$L = \mathrm{graph}(\Lambda) \subseteq \mathbb{T}M \times \mathfrak{g} \quad \text{is Dirac}$$

Cor: quasi-Poisson spaces $L \subseteq TM \times \mathcal{J}$ $\rightleftarrows$ Lagrangian structures

diff / integ.

$\mathcal{H}, \beta \xrightarrow{\phi} D$

$\downarrow$

M

$\mathrm{Lie}\,\phi$

$\ker \lambda_\beta \simeq \mathfrak{g} \times M$

Cor: quasi-Poisson spaces $\rightleftarrows$ Lagrangian structures

$L \subseteq \mathbb{T}M \times \mathfrak{g}$

diff / integ.

$$\begin{array}{ccc} \mathcal{H}, \beta & \longrightarrow & D \\ \downarrow\downarrow & \overset{\phi}{\longrightarrow} & \\ M & & \end{array}$$

$\mathrm{Lie}\,\phi$

$\ker \lambda_\beta \simeq \mathfrak{g} \times M$

$\rightsquigarrow$ description of integration in many cases ...

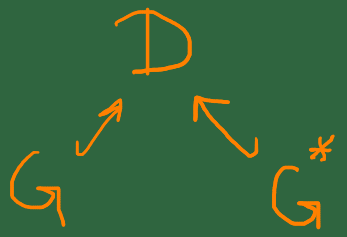
$$\begin{array}{c} \Sigma(M) \\ \downarrow\downarrow \\ G \curvearrowright (M, \pi) \end{array} \quad \rightsquigarrow \quad \mathcal{H} := \Sigma(M) \ltimes G \longrightarrow D$$

Iglesias-Fernandes, Li-Bland - Ševera

multiplicative moment map $\rightsquigarrow$ 2-shifted Lagrangian structure

Example: (G, π_G) Poisson Lie group

$\mathfrak{d} = \mathfrak{g} \oplus \mathfrak{g}^*$



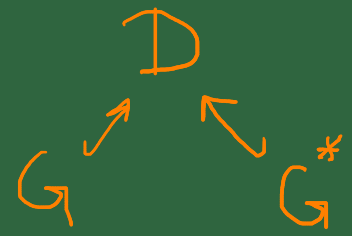
Poisson action: $G \curvearrowright G$ $\rightsquigarrow$ $\mathbb{L} \subseteq \mathbb{T}G \times \mathfrak{d}$

$a \longmapsto a g^{-1}$

integration?

Example: (G, π_G) Poisson Lie group

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Poisson action: $G \curvearrowright G$ $\quad a \longmapsto a g^{-1}$ $\rightsquigarrow$ $\mathbb{L} \subseteq \mathbb{T}G \times \mathfrak{d}$

integration?

(B., Iglesias, Lu)

Prop:

$$\mathcal{G}_D = \{(z, g_1, g_2, d) \in G^* \times G \times G \times D \mid z g_1 = g_2 d^{-1}\} \xrightarrow{\Phi} D$$

$\downarrow\downarrow$

G

$$(z, g_1, g, d) \cdot (z', g, g_2, d') = (z'z, g_1, g_2, d'd)$$

$$\beta = \frac{1}{2}\left(\langle \theta^L_{G,1}, \theta^r_D \rangle + \langle \theta^r_{G^*}, \theta^L_{G,2} \rangle \right) \quad \text{integrates } \mathbb{L}$$

④ Application

(G, π_G) Poisson Lie group

affine Dirac structure E on G : $(G, \pi_G) \times (G, E) \longrightarrow (G, E)$

Dirac map

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(G, π_G) Poisson Lie group

affine Dirac structure E on G : $(G, \pi_G) \times (G, E) \longrightarrow (G, E)$

Dirac map

$\mathfrak{d} \times G \xrightarrow{\tau}_{\sim} TG$, affine Dirac $\overset{\tau}{\rightleftharpoons}$ $\mathfrak{l} \subseteq \mathfrak{d}$ Lagrangian subalgebra

$$T^*G \hookrightarrow \mathfrak{d} \times G \twoheadrightarrow TG$$

dressing action

Action Courant algebroid is exact

④ Application

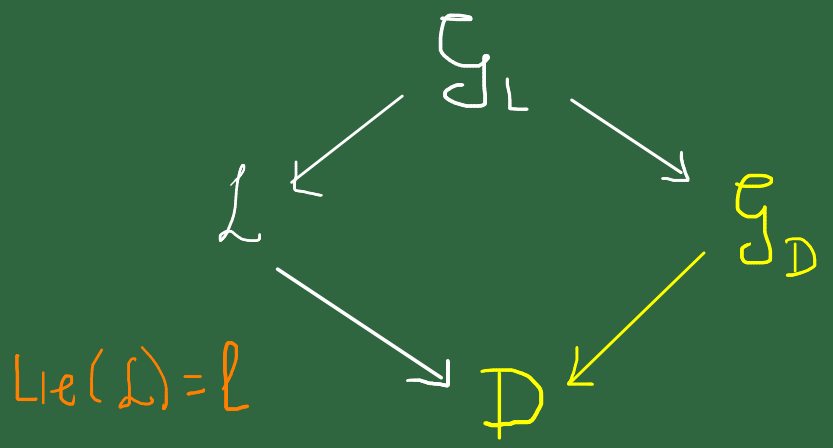
(G, π_G) Poisson Lie group $\quad \mathfrak{d} = \mathfrak{g} \oplus \mathfrak{g}^*$

affine Dirac structure E on G : $(G, \pi_G) \times (G, E) \longrightarrow (G, E)$

Dirac map

$\mathfrak{d} \times G \xrightarrow[\sim]{\tau} TG$ affine Dirac $\rightleftharpoons$ $\mathfrak{l} \subseteq \mathfrak{d}$ Lagrangian subalgebra

integration of affine Dirac :



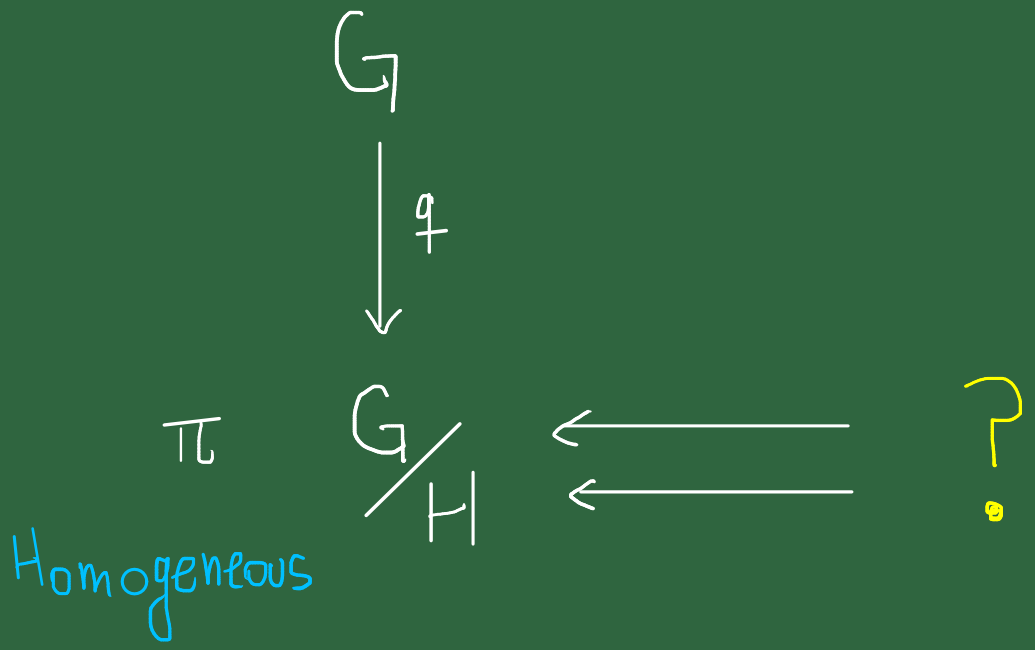
$$\mathcal{G}_L = \left\{ (z, g_1, g_2, l) \in \mathcal{G}^* \times G \times G \times \mathcal{L} \;\middle|\; z g_1 = g_2 l^{-1} \right\}$$

$$\omega_L = \frac{1}{2}\left(\langle \theta^L_{G,1}, \theta^r_L \rangle + \langle \theta^r_{G^*}, \theta^r_{G,2} \rangle \right)$$

Poisson homogeneous spaces

$H \subseteq G$ closed subgroup

(B., Iglesias, Lu)



Poisson homogeneous spaces

$H \subseteq G$ closed subgroup

$\mathfrak{l} \subseteq \mathfrak{g}$

$\mathfrak{g}^* \times G \xrightarrow{\tau}_{\sim} \Pi_G$

$\mathfrak{g}^* \cdot \pi$ affine

$(G, \{\cdot,\cdot\}_L)$ $\Longleftarrow$ $(\mathfrak{g}_L, \omega_L)$

$q \downarrow$

π Homogeneous $\quad (G/H, \{\cdot,\cdot\}) \Longleftarrow$?

Poisson homogeneous spaces

$H \subseteq G$ closed subgroup

$\mathfrak{g} \times G \xrightarrow[\sim]{\tau} TG$

$\mathfrak{l} \subseteq \mathfrak{g}$

$\mathfrak{g} \supset \pi$

affine

$G \rightrightarrows \mathfrak{g}_L$ ω_L $\circlearrowleft H \times H$

$q \downarrow$ $\quad\quad \vdots$

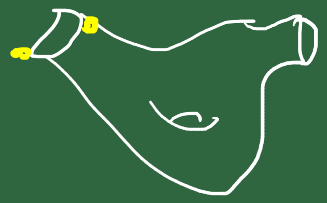
π $\quad G/H \rightrightarrows \mathfrak{g}_L/H \times H$

Homogeneous

symplectic groupoid !

Other applications:

- fusion of quasi-hamiltonian groupoids
  $\rightleftharpoons$ fusion of quasi-Poisson spaces
- integration of quasi-Poisson structures on representation spaces



Σ

$\hom(\pi_1(\Sigma, v), G)$

Li-Bland
Severa

Thank you!